

Homework 6: Improving Functions

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Improving analysis code by writing functions

Can you improve this analysis code?

```
library(bio3d)
s1 <- read.pdb("4AKE") # kinase with drug
s2 <- read.pdb("1AKE") # kinase no drug
s3 <- read.pdb("1E4Y") # kinase with drug

s1.chainA <- trim.pdb(s1, chain="A", elty="CA")
s2.chainA <- trim.pdb(s2, chain="A", elty="CA")
s3.chainA <- trim.pdb(s3, chain="A", elty="CA")

s1.b <- s1.chainA$atom
s2.b <- s2.chainA$atom
s3.b <- s3.chainA$atom

plotb3(s1.b, sse=s1.chainA, typ="l", ylab="Bfactor")
plotb3(s2.b, sse=s2.chainA, typ="l", ylab="Bfactor")
plotb3(s3.b, sse=s3.chainA, typ="l", ylab="Bfactor")
```

Q6. How would you generalize the original code above to work with any set of input protein structures?

Inputs: There are 3 inputs to this function: `pdb_id` (the PDB structure ID to analyze), `chain_id` (which protein chain to keep), and `atom_type` (which atom type to keep).

Purpose/How to Use: The function reads a protein structure from the Protein Data Bank (PDB) using an ID, trims the structure, extracts the corresponding B-factors, and plots the B-factors.

Output: The output of the function is a plot of the B-factors for the selected atoms.

```
library(bio3d)

plot_pdb_bfactors <- function(pdb_id, chain_id = "A", atom_type = "CA") {
  # read pdb id
  pdb <- read.pdb(pdb_id)

  # trim to desired structure
  chainA <- trim.pdb(pdb, chain = chain_id, eley = atom_type)

  # extract B-factors
  b <- chainA$atom$b

  # plot
  plotb3(b, sse = chainA, typ = "l", ylab = "Bfactor")
}
```

```
# example output:
plot_pdb_bfactors("4AKE")
```

Note: Accessing on-line PDB file

